Chapter 4-2: Experimental designs and Observational studies

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Experimental designs

Basic Terminologies of Experiments

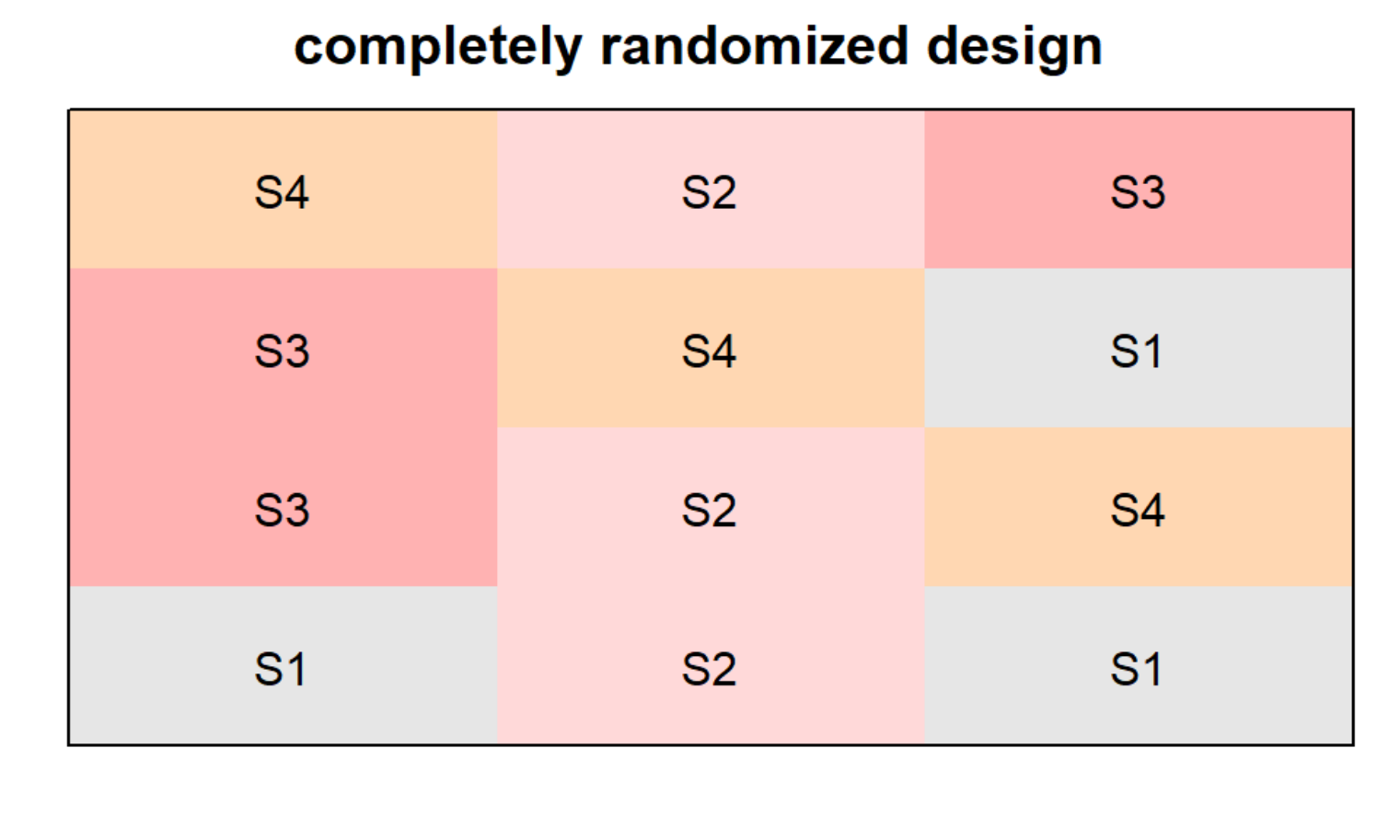
- *Experiment:* Investigation where treatments are applied to experimental units and responses are observed.
- *Treatment:* Condition or set of conditions applied to experimental units.
- *Response Variable:* Measured outcome used to assess treatment effects.
- *Experimental Unit:* The smallest division of material that can independently receive a treatment.
- *Observational Unit:* Unit on which the response is measured.

Principles of Design

- *Replication:* Apply treatment to multiple units to estimate variability.
- *Randomization:* Randomly assign treatments to reduce bias.
- *Blocking:* Group similar units; randomize within blocks.
- *Blinding:* Prevent subjects or researchers from knowing treatment assignment.
- *Control of Confounding:* Keep conditions consistent; use homogeneous units.
- *Control Group:* Compare treatments to placebo or standard.

Design 1: Completely Randomized Design (CRD)

Experimental units are randomly assigned to treatments.



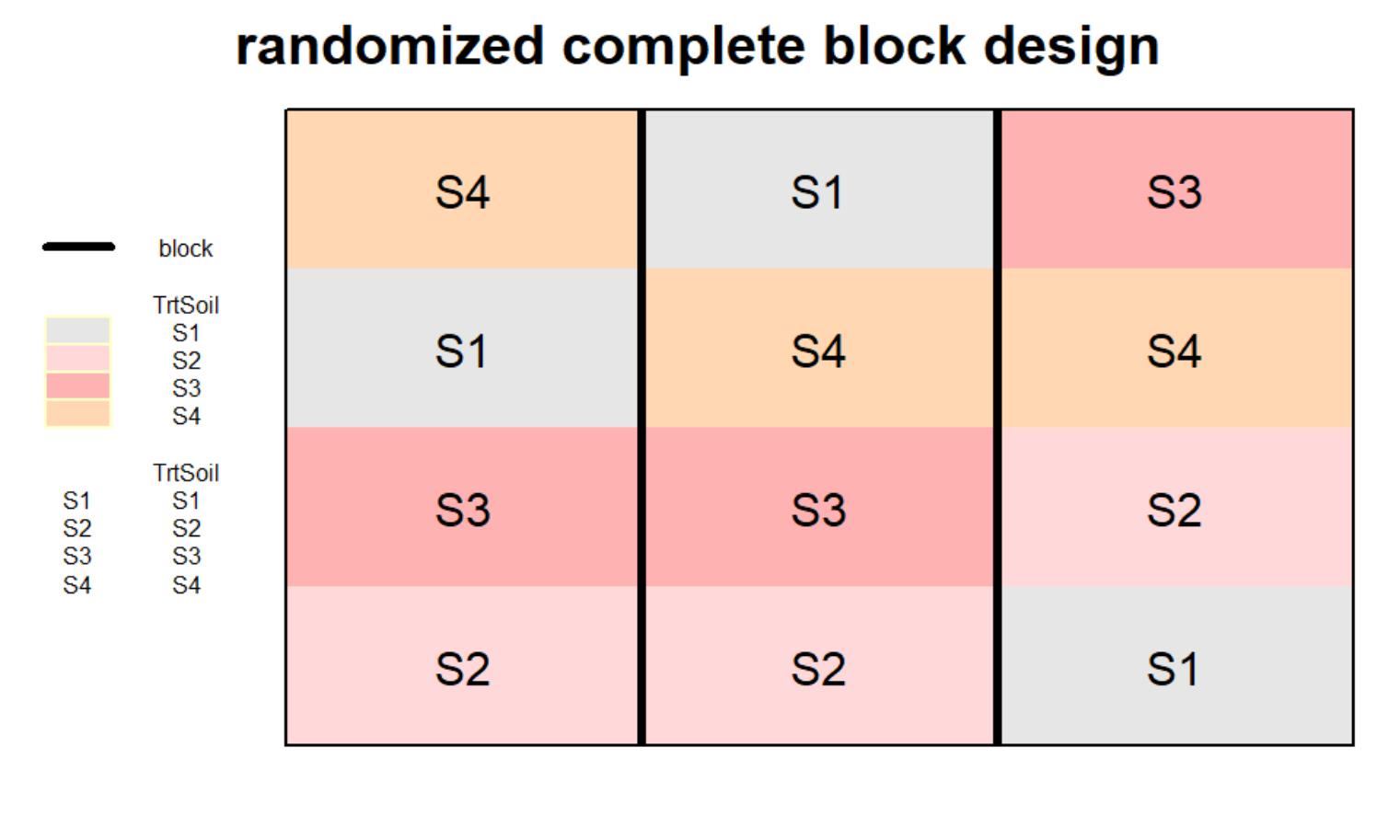
https://schmidtpaul.github.io/DSFAIR/DesigningExperiments.html

Example: Diet and Drug Effects in Pigs

- **Objective:** Study the effects of diet and drug type on weight gain in Yorkshire pigs.
- *Factors* and *Levels*:
  - Diet: Low fiber, High fiber.
  - Drug: D1, D2, D3.
- Design: 2×3 factorial, completely randomized.
- Experimental units: 12 pigs housed in individual pens.
- Procedure: Pigs injected weekly with assigned drug and fed assigned diet for 6 weeks.
- Response: Weight gained over the 6-week period.

Design 2: Randomized Complete Block Design (RCBD)

Randomized treatment assignments for each of multiple blocks.



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Example: Low Birthweight Babies

- **Objective:** Determine if family assistance program increases weight gain in first year.
- Two treatments(levels):
  - Printed information only.
  - Printed information + nurse visits.
- Design: RCBD with 3 blocks(weight classes) and 2 treatments.
- Blocking factor: 3 weight classes; 30 babies per class.
- Experimental units: 90 babies.
- Procedure: Within each weight class, 15 babies randomly assigned to each treatment.
- Response: Weight gain during the first year of life.

Example: Low Birthweight Babies

Weight Class (Block)	Treatment 1	Treatment 2	Total Units
Lightest (Class 1)	15	15	30
Medium (Class 2)	15	15	30
Heaviest (Class 3)	15	15	30
Total	**45**	**45**	**90**

Observasional studies

What are Observational Studies?

- Sometimes, treatments cannot be assigned to experimental units.
- Example: studying the effect of smoking on cancer in humans.
- Data are gathered by observing subjects as they naturally exist.
- These are called *observational studies*, not experiments.
- **Key limitation:** We can study associations, not causation.

Bias Resulting from Study Design

- *Selection bias*: systematic error in the selection or retention of participants
- *Information bias:*
  - *Non-response bias:* sampled units fail to respond.
  - *Recall bias:* inaccurate or untruthful answers.

Cohort vs. Case–Control Study

Cohort Study

- Observational, usually **prospective**.
- Start with groups defined by **exposure**.
- Follow over time for outcome.
- *Example:* Track smokers vs. non-smokers for lung cancer.

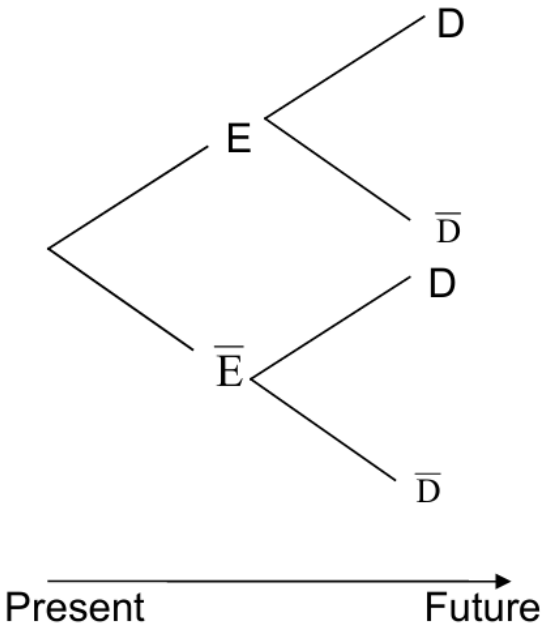
Case–Control Study

- Observational, **retrospective**.
- Start with groups defined by **outcome**.
- Look back for prior exposure.
- *Example:* Compare smoking history of lung cancer cases and controls.

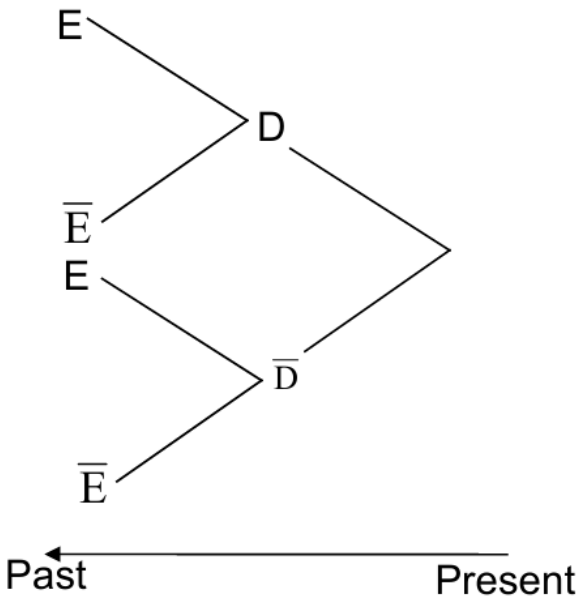
Cohort: exposure $\rightarrow$ outcome; Case–control: outcome $\rightarrow$ exposure.

Cohort vs. Case–Control Study

Cohort design:



Case-control design:



Cohort Study Example: Diet and Heart Disease

- **Objective:** Examine the relationship between diet and heart disease.
- **Design:** Prospective, *cohort design*.
  - 10,000 adults surveyed about dietary habits at baseline.
- **No random assignment**: diet and lifestyle were self-selected.
- **Explanatory variable:** Dietary pattern.
- **Response:** Development of heart disease during follow-up.

	Heart Disease	**No Disease**	**Total**
High-fat Diet	150	6,350	6,500
Low-fat Diet	50	3,450	3,500
Total	200	9,800	10,000

Case-Control Study Example: Smoking and Lung Cancer

- **Objective:** Investigate the relationship between smoking and lung cancer.
- **Design:** Retrospective, *case–control design*.
  - Cases: Individuals diagnosed with lung cancer.
  - Controls: Individuals without lung cancer.
- **No random assignment**: smoking behavior was self-selected.
- **Response:** Presence or absence of lung cancer.
- **Explanatory variable:** Smoking history.

	Lung Cancer	**No Cancer**	**Total**
Smoker	68	24	92
Non-smoker	12	46	58
Total	80	70	150